1A-10

Letting A, B, C, D be the four sides, then the vectors are suitably oriented, we have

$$A + B = C + D$$

The vector from the midpoint of A to the midpoint of C is

$$\frac{1}{2}C - \frac{1}{2}A$$

Similarly, the vector joining the midpoint of the other two sides is

$$\frac{1}{2}B - \frac{1}{2}D$$

$$A + B = C + D$$

$$C - A = B - D$$

$$\frac{1}{2}(C - A) = \frac{1}{2}(B - D)$$

Thus two opposite sides are equal and parallel, which shows the figure is a parallelogram.

1A-11

Let the four vertices be O, P, Q, R, with X on PR and Y on OQ.

$$OX = OP + PX = OP + \frac{1}{2}PR$$

$$= OP + \frac{1}{2}(OR - OP)$$

$$= \frac{1}{2}(OR + OP)$$

$$= \frac{1}{2}OQ = OY$$

Therefore $X = Y$.

1B Dot Product

1B-1

(a)

$$\cos\theta = \frac{A \cdot B}{|A||B|} = \frac{4+2}{\sqrt{2} \cdot 6} = \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4}$$

(b)

$$\cos\theta = \frac{3}{\sqrt{6}\cdot\sqrt{6}} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

1B: Dot Product
1B-2

$$A \cdot B = c - 4$$

(a) Orthogonal if

$$c - 4 = 0 \quad \Rightarrow \quad c = 4$$

(b)

$$\cos\theta = \frac{c-4}{\sqrt{c^2+5}\sqrt{6}}$$

The angle is acute if

$$\cos\theta > 0 \Rightarrow c > 4$$

1B-3

Place the cube in the first octant so the origin is at one corner. $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the three edges.

Longest diagonal:

$$PQ = \mathbf{i} + \mathbf{j} + \mathbf{k}$$

Face diagonal:

$$PR = \mathbf{i} + \mathbf{j}$$

(a)

$$\cos\theta = \frac{PQ \cdot PR}{|PQ||PR|} = \frac{2}{\sqrt{3}\sqrt{2}}$$

$$\theta = \cos^{-1}\left(\frac{2}{\sqrt{6}}\right)$$

(b)

$$\cos\theta = \frac{PQ \cdot \mathbf{i}}{|PQ||\mathbf{i}|} = \frac{1}{\sqrt{3}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

1B-4

$$QP = (a, 0, -2), \qquad QR = (a, -2, 2)$$

(a)

$$QP \cdot QR = a^2 - 4$$

Right angle if

$$a^2 - 4 = 0 \Rightarrow a = \pm 2$$

(b)

$$\cos\theta = \frac{a^2 - 4}{\sqrt{a^2+4}\sqrt{a^2+8}}$$

Angle is acute if

$$\cos\theta > 0 \Rightarrow a^2 > 4$$

$$|a| > 2$$

1B-5
(a)

$$F \cdot U = -\frac{1}{\sqrt{3}}$$

Since $b = \text{dir } A = \frac{A}{|A|}$, therefore

$$F \cdot U = \frac{4}{7}$$

1B-6
After dividing by $|OP|$ the equation becomes

$$\cos\theta = c$$

where θ is the angle between OP and U.
Let

$$\theta_0 = \cos^{-1}(c)$$

Then the locus is the nappe of a right circular cone with axis in the direction U and vertex angle $2\theta_0$.

In particular:
(a) A plane if

$$\theta_0 = \frac{\pi}{2} \quad (c = 0)$$

(b) A ray if

$$\theta_0 = 0 \quad (c = 1)$$

(c) The locus is the origin if

$$|c| > 1$$

1B-7
(a)

$$|i'| = |j'| = \frac{\sqrt{2}}{\sqrt{2}} = 1$$

A picture shows the system is right–handed.

(b)

$$A \cdot i' = -\frac{1}{\sqrt{2}}, \qquad A \cdot j' = -\frac{5}{\sqrt{2}}$$

Since they are perpendicular unit vectors,

$$A = -i' + j'$$

(c)

$$i = \frac{i' - j'}{\sqrt{2}}, \qquad j = \frac{i' + j'}{\sqrt{2}}$$

Thus

$$A = 2i - 3j$$

1B-8

a) Check that each length is 1, and the three dot products $i' \cdot j'$, $i' \cdot k'$, $j' \cdot k'$ are 0. Make a sketch to check right-handedness.

b) $A \cdot i' = \sqrt{3}, \quad A \cdot j' = 0, \quad A \cdot k' = \sqrt{6}$

$$A = \sqrt{3}\, i' + \sqrt{6}\, k'$$

1B-9

Let $u = \text{dir } A$, then the vector u-component of B is $(B \cdot u)u$. Subtracting it off gives a vector perpendicular to u (and therefore also to A), thus

$$B = (B \cdot u)u + \big(B - (B \cdot u)u\big)$$

or in terms of A, remembering that $|A|^2 = A \cdot A$

$$B = \frac{B \cdot A}{A \cdot A} A + \left(B - \frac{B \cdot A}{A \cdot A} A\right)$$

1B-10

Let two adjacent edges of the parallelogram be the vectors A and B. Then the two diagonals are $A + B$ and $A - B$.

Remembering that for any vector C we have

$$C \cdot C = |C|^2$$

$$(A + B) \cdot (A + B) = (A - B) \cdot (A - B)$$

$$(A \cdot A) + 2(A \cdot B) + (B \cdot B) = (A \cdot A) - 2(A \cdot B) + (B \cdot B)$$

$$A \cdot B = 0$$

which says the two sides are perpendicular.

PB-11
Using the notation of the previous exercise, we have successively

$$(A+B)\cdot(A-B)=A\cdot A-B\cdot B$$

Therefore

$$(A+B)\cdot(A-B)=0 \iff A\cdot A=B\cdot B$$

The diagonals are perpendicular if and only if two adjacent edges have equal lengths. In other words, the parallelogram is a rhombus.

1B-12
Let O be the center of the semicircle and Q and R the two ends of the diameter, and P the vertex of the inscribed angle.
Set

$$A=\overrightarrow{QO}=\overrightarrow{OR}, \qquad B=\overrightarrow{OP}$$

Then

$$|A|=|B|.$$

The angle sides are

$$QP=A+B \qquad \text{and} \qquad PR=A-B$$

They are perpendicular since

$$(A+B)\cdot(A-B)=A\cdot A-B\cdot B=0$$

because $|A|=|B|$.

1B-13
The unit vectors are

$$u_i=(\cos\theta_i,\sin\theta_i), \qquad i=1,2$$

If $\theta_2-\theta_1$ is the angle between them, we use the geometric definition of the dot product:

$$\cos(\theta_2-\theta_1)=\frac{u_1\cdot u_2}{|u_1||u_2|}$$

$$=\cos\theta_1\cos\theta_2+\sin\theta_1\sin\theta_2$$

According to the formula for evaluating the dot product in terms of components.
1B-4
Let the coterminal vectors A and B represent two sides of the triangle, and let θ be the included angle. Suitably directed, the third side is

$$C=A-B$$

$$|C|^2 = (A - B) \cdot (A - B) = A \cdot A + B \cdot B - 2A \cdot B$$

$$= |A|^2 + |B|^2 - 2|A||B| \cos\theta$$

1C Determinants

1C-1

a)

$$\begin{vmatrix} 1 & 4 \\ 2 & -1 \end{vmatrix} = (1)(-1) - (4)(2) = -1 - 8 = -9$$

b)

$$\begin{vmatrix} 3 & -4 \\ -1 & -2 \end{vmatrix} = (3)(-2) - (-4)(-1) = -6 - 4 = -10$$

1C-2

$$\begin{vmatrix} -1 & 0 & 4 \\ 1 & 2 & 2 \\ 3 & -2 & -1 \end{vmatrix}$$

a) By the cofactors of row one

$$= -1 \begin{vmatrix} 2 & 2 \\ -2 & -1 \end{vmatrix} - 0 \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} + 4 \begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix}$$

$$= -34$$

b) By the cofactors of column one

$$= -1 \begin{vmatrix} 2 & 2 \\ -2 & -1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 4 \\ -2 & -1 \end{vmatrix} + 3 \begin{vmatrix} 0 & 4 \\ 2 & 2 \end{vmatrix}$$

$$= -34$$